

Taxation and the Welfare State: Solutions to Problem Set 9

Arthur Seibold, LMU

Summer 2026

Question 1)

a.

The unemployment rate in Germany was around 5% in the early 1990s and rose to a peak of more than 11% in 2005. This was followed by a large decline in unemployment until 2019, when the unemployment rate hit a low of around 3%. Since then, the unemployment rate has slightly increased to 2025.

b.

The German unemployment rate reflects both long-term economic trends and short-term business cycle fluctuations. In the late 1990s and early 2000s, the German economy was stagnant, which led to high unemployment. Between 2005 and 2019, the German economy saw steady GDP growth, which led to historically low unemployment. Major economic crises have had surprisingly little impact on the German unemployment rate. For instance, unemployment only saw small peaks during the financial crises and the Covid-19 pandemic.

c.

Of course we don't know how unemployment will evolve in the next 10 years, but we can speculate. One important factor is the low economic growth Germany has seen over the last 5 years – this has not yet translated to significant unemployment, but it may in the future. Another important factor is technological change: there are fears that AI could replace many jobs, which may lead to unemployment – we don't know yet if this will happen.

Question 2)

a.

In the first-best scenario, the government solves

$$\max_{b, \tau, e} \pi(e) u(\underbrace{w - \tau}_{=c_e}) + [1 - \pi(e)] u(\underbrace{b}_{=c_u}) - e$$

subject to the government budget constraint

$$\pi(e)\tau - [1 - \pi(e)]b = 0$$

The government budget constraint implies

$$\tau(b) = \frac{1 - \pi(e)}{\pi(e)} b \quad (1)$$

such that we can re-write the government's problem as

$$\max_{b, e} \pi(e) u \left(w - \frac{1 - \pi(e)}{\pi(e)} b \right) + [1 - \pi(e)] u(b) - e$$

The first-order condition w.r.t. e is

$$\frac{\partial}{\partial e} = \pi'(e) [u(c_e) - u(c_u)] - 1 + \pi(e) u'(c_e) \frac{\pi'(e)}{[\pi(e)]^2} b = 0$$

which implies

$$\underbrace{\pi'(e) [u(c_e) - u(c_u)]}_{\text{PMB of search}} + \underbrace{\frac{\pi'(e)}{\pi(e)} u'(c_e) b}_{\text{fiscal externality}} = \underbrace{1}_{\text{PMC of search}}$$

The first-order condition w.r.t. b is

$$\frac{\partial}{\partial b} = \pi(e) u'(c_e) \left[-\frac{1 - \pi(e)}{\pi(e)} \right] + [1 - \pi(e)] u'(c_u) = 0$$

which implies $u'(c_e) = u'(c_u)$ or $c_e = c_u$ and thus full insurance.

Unfortunately the first-best solution is not attainable in practice as the government cannot choose (observe and/or enforce) e .

b.

The agent's problem is

$$\max_e \pi(e) u(w - \tau) + [1 - \pi(e)] u(b) - e$$

Note that the agent takes the policy b, τ as given and only chooses his search effort e . The individual optimum is characterized by the FOC

$$\frac{\partial}{\partial e} = \pi'(e) [u(c_e) - u(c_u)] - 1 = 0$$

or

$$\underbrace{\pi'(e) [u(c_e) - u(c_u)]}_{\text{PMB of search}} = \underbrace{1}_{\text{PMC of search}} \quad (2)$$

Hence, the agent's optimal search effort differs from the first-best solution in part a., as the agent does not take into account the fiscal externality.

c.

For the second-best solution, the government solves

$$\max_{b, \tau} \pi(e)u(w - \tau) + (1 - \pi(e))u(b) - e$$

subject to equations (1) and (2), which is equivalent to solving

$$\max_b \pi(e(b))u(w - \tau(b)) + [1 - \pi(e(b))]u(b) - e(b)$$

where the function $\tau(b)$ is implicitly defined by equation (1) and $e(b)$ is implicitly defined by equation (2). We can take the FOC w.r.t. b and with some steps of algebra we obtain the Baily-Chetty formula:

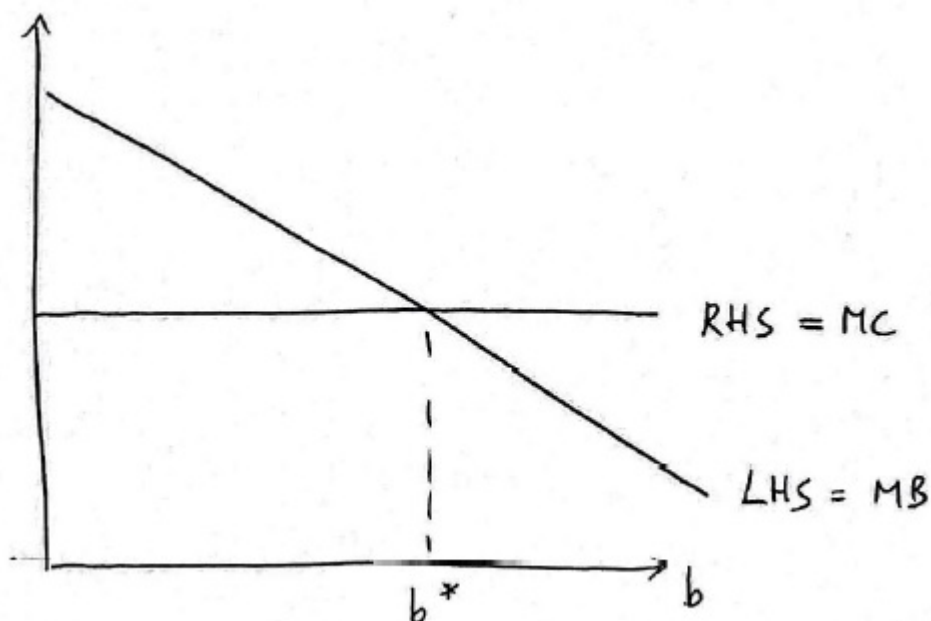
$$\gamma \frac{\Delta c}{c} \approx \frac{\varepsilon_{1-\pi(e), b}}{\pi(e)}$$

The left hand side is the marginal benefit of increasing b in terms of consumption smoothing, the right hand side represents the marginal cost in terms of moral hazard.

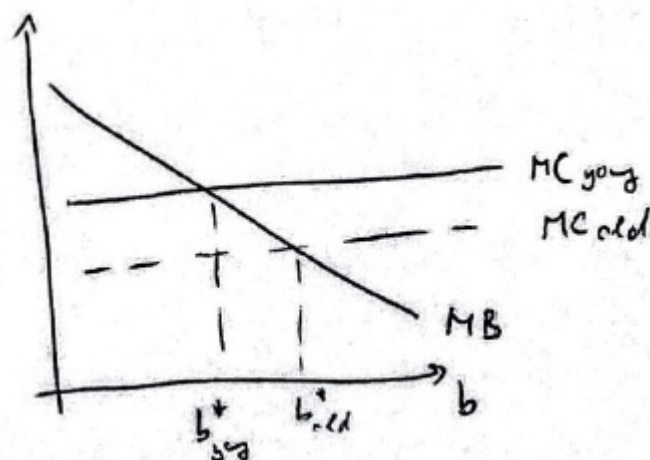
[Note: You can try to take the FOC w.r.t. b and derive the formula yourself, but you are not required to do this for the exam.]

d.

We can draw a graphical representation of the Baily-Chetty formula. In each of the scenarios below, we discuss how either the LHS or the RHS of the formula may change, and use the graph to see how this would change the optimal level of benefits b^* . The LHS is decreasing in b , as Δc gets smaller when b increases. We can assume the RHS is constant in b .



i. Older workers may find it more difficult to find a job, such that their search effort affects their job finding probability less. This would imply a lower elasticity and thus a lower marginal cost (RHS). Shifting the MC curve downwards, we can see that this would imply a higher b^* for older workers (see figure below).



ii. When workers have savings, this would imply a smaller Δc , as workers can smoothe their consumption by using their savings when unemployed. Such self-insurance implies a smaller insurance value of unemployment benefits (LHS of formula). When the MB curve is shifted down, we get a lower optimal b^* .

iii. When job seekers are over-optimistic and they under-respond to changes in b , this would imply that the elasticity is lower. With a lower MC curve, we get a higher optimal b^* .

Question 3)

a.

The econometric problem of Gruber (1997) is:

$$\left(\frac{c_e - c_u}{c_e} \right)_{i,s,t} = \beta_1 + \beta_2 \left(\frac{b}{w} \right)_{i,s,t} + X'_{i,s,t} \gamma + \delta_s + \zeta_t + \epsilon_{i,s,t}$$

where $X_{i,s,t}$: controls, δ_s : state fixed effect, ζ_t : time fixed effect and $\epsilon_{i,s,t}$: error term. The subscripts are i : individual, s : state, and t : time.

There are two potential sources of endogeneity of the UI replacement rate b/w :

1. Individual earnings before unemployment mechanically affect b/w , and are also likely to affect the outcome Δc
2. Policy endogeneity: states may change b/w in response to demands by citizens, which may depend on their consumption smoothing opportunities

Hence, the equation cannot just be estimated via OLS.

b.

Simulated IV approach. Gruber (1997) simulates state-level UI benefits for each period on a constant sample of individuals. Then the actual b/w is instrumented with the simulated b/w . Hence, variation in UI generosity across states and time arising from policy is used (and not the variation due to individual characteristics within state-time). The usual identification assumptions for IV must hold:

- First stage: simulated b/w must be correlated with actual b/w .
- Exclusion restriction: state-time variation in b/w is only related to consumption via the causal effect of interest

The first-stage assumption is likely to hold. The exclusion restriction, on the other hand, requires that there is no policy endogeneity. Hence, the simulated IV approach eliminates the individual endogeneity problem 1 from part a., but problem 2 remains a concern.

c.

Food consumption may not vary with income to the same extent as other types of consumption. For instance, since food is a necessity, other types of consumption may respond more to income changes, such that the insurance value is underestimated when using food consumption. On the other hand, since expenditure on food can be more easily substituted by home production when unemployed, food consumption measured as expenditure may vary more than other types which would lead to overestimation of the insurance value.